S. B. JAIN INSTITUTE OF TECHNOLOGY, MANAGEMENT & RESEARCH, NAGPUR.

Practical No. 9

Aim: Evaluate Bayes Rules for Rain predication from given scenario.

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Academic Session:

Date of Performance:

Date of Submission:

AIM: Evaluate Bayes Rules for Rain predication from given scenario.

OBJECTIVE/EXPECTED LEARNING OUTCOME:

The objectives and expected learning outcome of this practical are:

- To be able to understand the concept of Bayes theorem.
- To be able to solve problem based on Bayes theorem.

THEORY:

Bayes Theorem: - Bayes' Theorem states that the conditional probability of an event, based on the occurrence of another event, is equal to the likelihood of the second event given the first event multiplied by the probability of the first event.

Bayes theorem is used to find the reverse probabilities if we know the conditional probability of an event.

Bayes' Theorem states the following for any two events A and B:

$$P(A|B) = \frac{P(B|A)\,P(A)}{P(B)}$$

where:

· P(A|B): The probability of event A, given event B has occurred.

· P(B|A): The probability of event B, given event A has occurred.

· P(A): The probability of event A.

· P(B): The probability of event B.

Terms Related to Bayes' Theorem

After learning about Bayes theorem in detail, let us understand some important terms related to the concepts we covered in the formula and derivation.

- **Hypotheses:** Hypotheses refer to possible events or outcomes in the sample space; they are denoted as E1, E2, …, En.

Each hypothesis represents a distinct scenario that could explain an observed event.

- **Priori Probability:** Priori Probability P(Ei) is the initial probability of an event occurring before any new data is taken into account.

  It reflects existing knowledge or assumptions about the event.

Example: The probability of a person having a disease before taking a test.

- **Posterior Probability:** Posterior probability (P(Ei|A) is the updated probability of an event after considering new information.

It is derived using the Bayes Theorem.

Example: The probability of having a disease given a positive test result.

- **Conditional Probability:** The probability of an event A based on the occurrence of another event B is termed conditional Probability.

  It is denoted as P(A|B) and represents the probability of A when event B has already happened.

- **Joint Probability:** When the probability of two or more events occurring together and at the same time is measured, it is marked as Joint Probability.

  For two events A and B, it is denoted by joint probability is denoted as P(A∩B).

- **Random Variables:**

  Real-valued variables whose possible values are determined by random experiments are called random variables.

  The probability of finding such variables is the experimental probability.

Rain Prediction Example: Suppose the probability of the weather being cloudy is 40%. Also suppose the probability of rain on a given day is 20%.

Also suppose the probability of clouds on a rainy day is 85%.

If it's cloudy outside on a given day, what is the probability that it will rain that day?

Solution:

· P(cloudy) = 0.40
· P(rain) = 0.20
· P (cloudy | rain) = 0.85
· Thus, we can calculate:
· (rain | cloudy) = P(rain) * P (cloudy | rain) / P(cloudy)
· P (rain | cloudy) = 0.20 * 0.85 / 0.40
· P (rain | cloudy) = 0.425
· If it's cloudy outside on a given day, the probability that it will rain that day is 42.5%.

PROGRAM CODE:

weather.numeric - Notepad

File Edit Format View Help

```
@relation weather

@attribute outlook {sunny, overcast, rainy}
@attribute temperature numeric
@attribute humidity numeric
@attribute windy {TRUE, FALSE}
@attribute play {yes, no}

@data
sunny,85,85,FALSE,no
sunny,80,90,TRUE,no
overcast,83,86,FALSE,yes
rainy,70,96,FALSE,yes
rainy,68,80,FALSE,yes
rainy,65,70,TRUE,no
overcast,64,65,TRUE,yes
sunny,72,95,FALSE,no
sunny,69,70,FALSE,yes
rainy,75,80,FALSE,yes
sunny,75,70,TRUE,yes
overcast,72,90,TRUE,yes
overcast,81,75,FALSE,yes
rainy,71,91,TRUE,no
```

Weka Explorer

Preprocess | Classify | Cluster | Associate | Select attributes | Visualize

Open file... | Open URL... | Open DB... | Generate... | Undo | Edit... | Save...

Filter

Choose **None** | Apply | Stop

Current relation
Relation: weather
Instances: 14
Attributes: 5
Sum of weights: 14

Attributes

All | None | Invert | Pattern

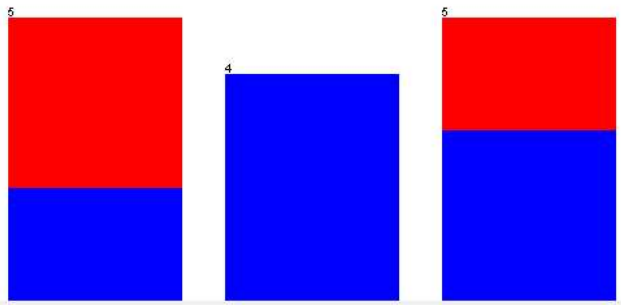
No.	Name
1	outlook
2	temperature
3	humidity
4	windy
5	play

Remove

Selected attribute
Name: outlook
Missing: 0 (0%)
Distinct: 3
Type: Nominal
Unique: 0 (0%)

No.	Label	Count	Weight
1	sunny	5	5
2	overcast	4	4
3	rainy	5	5

Class: play (Nom) | Visualize All



Status

Weka Explorer

Preprocess | Classify | Cluster | Associate | Select attributes | Visualize

Classifier

Choose **ZeroR**

Test options
- Use training set
- Supplied test set — Set...
- Cross-validation — Folds 10
- Percentage split — % 66

More options...

(Nom) play

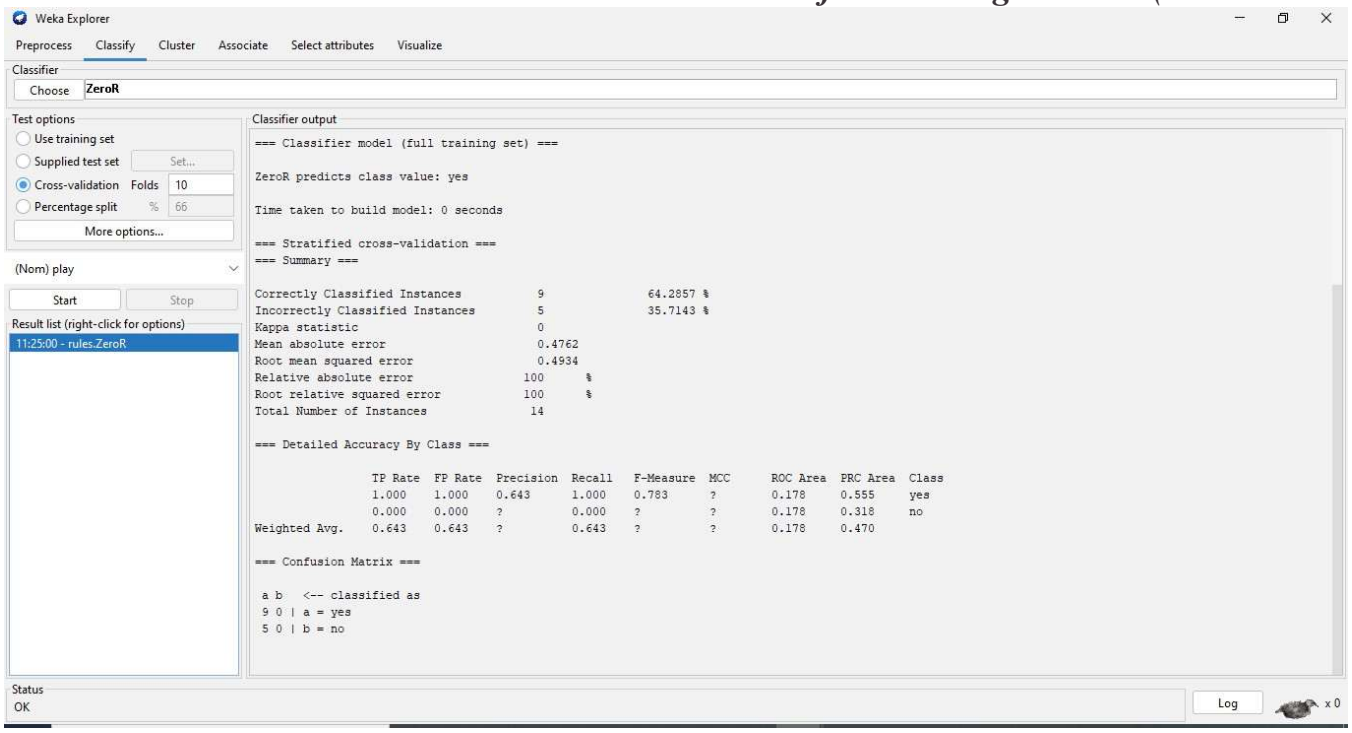
Start | Stop

Result list (right-click for options)
11:25:00 - rules.ZeroR

Classifier output

```
=== Run information ===

Scheme:       weka.classifiers.rules.ZeroR
Relation:     weather
Instances:    14
Attributes:   5
              outlook
              temperature
              humidity
              windy
              play
Test mode:    10-fold cross-validation

=== Classifier model (full training set) ===

ZeroR predicts class value: yes

Time taken to build model: 0 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances           9               64.2857 %
Incorrectly Classified Instances         5               35.7143 %
Kappa statistic                          0
Mean absolute error                      0.4762
Root mean squared error                  0.4934
Relative absolute error                100      %
Root relative squared error            100      %
Total Number of Instances               14

=== Detailed Accuracy By Class ===
```

Status



CONCLUSION:

DISCUSSION QUESTIONS:

1. What is the correct formula for Bayes Theorem?

2. Which algorithm uses Bayes theorem?

3. When can we use Bayes theorem?

4. How is Bayes theorem used in real life?

REFERENCES:

- *https://www.geeksforgeeks.org/maths/bayes-theorem/*
- *https://discovery.cs.illinois.edu/learn/Prediction-and-Probability/Bayes-Theorem/*
- *https://blogs.cornell.edu/info2040/2021/11/05/sunny-with-a-chance-of-rain-using-bayes-theorem-to-predict-the-weather/*